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RESEARCH ARTICLE

Trust and Ethics in Chatbots: A Framework for Responsible AI Interactions

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ABSTRACT The increasing deployment of chatbots across sectors such as education, healthcare, and customer service has raised the need for user interactions with the chatbot be safe, fair, responsible and respectful. These expectations from a chatbot could be collectively named as ethical behaviour or ethical concerns in AI applications. Human beings by default are expected to possess moral values, ethics and fairness. Indeed, critical AI applications should also be aligned with these qualities as they deal with real time scenarios and people. Ethical concerns include human agency and oversight, technical robustness and safety, transparency, non-discrimination and fairness, accountability, societal well-being, privacy and data governance. This paper explores the ethical implications of chatbots that exhibit varying responses due to fluctuations in dataset intents and patterns. Inconsistent behaviour can undermine user trust, introduce bias, and compromise decision-making, especially when the chatbot is used in sensitive domains. The study investigates how changes in the structure, labelling, or balance of intent patterns influence chatbot performance, potentially leading to unequal treatment of users or miscommunication on a university chatbot. Ethical principles in context to correctly classifying intent proves that chatbot is technically robust with a accuracy rate of 98% and a safety score of 87.65%, proves diversity and fairness with a fairness score of 84.5%, and existing frame works IBM AIF Fairness 360 and Microsoft RAI Toolkit is used for comparison with accuracy of 99.5% and 99.12% respectively and data governance achieved with a data quality score of 96.8% and contributes Societal and environmental well being with a sustainability score 84%. Recommendations are proposed for implementing auditability, explainability, and robust validation processes to ensure that chatbot systems remain ethically compliant even under dynamic or evolving datasets. The research underscores the need for continuous monitoring and ethical auditing mechanisms to safeguard against unintended harms and promote responsible AI deployment. To add on value two additional domains healthcare and customer service chatbot are used for comparison.

INDEX TERMS Artificial intelligence, chatbot, trustworthiness, ethical principles, sustainability.

I. INTRODUCTION

Today technology is inseparable from Artificial Intelligence(AI). Artificial Intelligence is ability of machines or computers to perform certain tasks that otherwise requires human intelligence. Artificial intelligent (AI) systems have increasingly embedded themselves in everyday digital interactions, chatbots have emerged as one of the prominent tools for communication, support, and decision-making. From customer service platforms to mental health apps, chatbots are now entrusted with sensitive tasks and data. However, with growing systems in capability and autonomy, a pressing question arises: Can they be trusted? Trust [1] is a critical component in the success and adoption of chatbot technologies. While performance [25] and accuracy are often emphasized, the ethical dimensions of chatbot

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trustworthiness remain underexplored. Ethical factors—such as transparency, fairness, accountability, privacy, and user autonomy—play prominent role in shaping the user trust. A chatbot that provides accurate information but manipulates conversations, conceals its identity, or mishandles personal data cannot be considered truly trustworthy. This study examines the ethical foundations of chatbot trustworthiness, aiming to identify the principles and practices necessary for building ethically responsible conversational agents. By analysing key ethical concerns [2] and proposing guidelines for their integration into chatbot design, this work acts as a bridge to reduce the gap between technical performance and moral responsibility. Ultimately, fostering trust in chatbots requires not only intelligent algorithms but also a deep commitment to ethical values [3], [24], [29]. Our proposed model categorizes ethical principles with their aspect technically implementable or not, particularly with university chatbot. Every ethical aspect may behave differently with different dataset; proposed model provides the comparison of different datasets with same ethical aspects.

II. BACKGROUND STUDY

The development of chatbots has seen rapid advancement in recent years, due to growing development in Natural language processing (NLP), Machine learning(ML), and Human-computer interaction (HCI). From early rule-based systems like ELIZA [4] to today's sophisticated AI-driven assistants OpenAI's ChatGPT and Google's Bard, chatbots have evolved from simple scripted responders to complex systems capable of maintaining context-rich conversations. With their increasing ubiquity in sectors such as health sector, finance business, education, and customer service now chatbots are not only expected to perform efficiently but also to act responsibly. This shift has contributed to the concept of trustworthiness—a multidimensional construct involving credibility, reliability, and integrity [5].

Trust in chatbots has traditionally been evaluated through technical measures such as response accuracy, system availability, and task completion rates. However, researchers and ethicists have begun to emphasize that these metrics are insufficient on their own. There are seven fundamental ethical principles as reflected in the European AI Regulations, taken from the European Commission's 2019 Guidelines as shown in FIGURE 1. Ethical principles include human agency and oversight, technical robustness and safety, transparency, non-discrimination and fairness, privacy and data governance, accountability, societal well-being. Ethical concerns, such transparency (whether users know they are speaking to a bot), fairness (avoiding biased or discriminatory responses), privacy (protecting user data), and accountability (who is responsible when things go wrong), are now recognized as essential factors influencing trust [6], [15], [16], [37].

Research on human conversation with computers suggests that users more likely trust systems which are explainable, predictable, and respectful of their autonomy. Furthermore, recent work in AI ethics has proposed design frameworks that embed moral values into AI systems, advocating for ''ethical-by-design'' approaches [7]. Despite these advancements, there remains a gap in systematically integrating ethical considerations into the design and evaluation of chatbot systems. Many commercial bots still lack clear ethical guidelines, potentially leading to erosion of user trust, especially in high-stakes domains like mental health or legal advice. This background highlights the necessity of examining chatbot trust not just from a functional standpoint but through the lens of ethics. Addressing this need forms the foundation of the study.

III. RELATED WORKS

The intersection of trust and ethics in conversational AI has become a growing area of academic inquiry, with researchers examining how design choices influence user perception and system reliability. Existing literature can be broadly categorized into three main areas: trust in chatbot interaction, ethical AI frameworks [1], and trust-enhancing design strategies.

1. Trust in Chatbot Interaction: Due to advancements in natural language processing and artificial intelligence, we are surrounded by a variety of conversational agents, or chatbots. Convenient services like food ordering, stock recommendations, and fund diagnostics can be offered by these chatbots. How to convince people that the chatbots are trustworthy is still unclear, though. In order to foster trust between human and computer systems, certain design principles are investigated in this study [8]. The study's main conclusions are that there are two ways to gauge perceptions of trust. We select whom to trust with respect to what circumstances, and the choice is based on the good reasons, constituting evidence towards trust-worthiness, according to the first component cognitive-based trust, is represented by the chatbot. Affect-based trust, the second component, is created by emotional ties between people that transcend a typical business or professional relationship. [9]

Four popular banking chatbots users in India—SBI Intelligent Assistant, HDFC Bank's Electronic Virtual Assistant, ICICI Bank's iPal, and Axis Aha—were considered for an online survey. According to the findings, 38.6% difference in banking chatbot trust could be explained by the hypothesized antecedents, with the exception of interface, design, and technological fear concerns. Additionally, when it comes to behavioural outcomes, chatbot trust could account for 13.6% difference in user happiness, 11.4% difference in behavioural intention, and 9.9% difference in customer attitude. The report gives managers useful information about how to use chatbot trust to boost consumer engagement with their company [10]. Trust in chatbot is domain specific and the efficiency of the application is specific to the user. In banking chatbots or healthcare the efficiency is expected to be 100% where no compromise is acceptable, unlike chatbots for customer service etc. Proposed model focuses on university chatbot, where trust measures are very clear based on seven ethical principles recommended by European commission, unlike trust measured only by consumer engagement with the chatbot where trust parameters are uncertain.

2. Ethical Frameworks in AI: The prominence of ethics in AI system design has been emphasized by major institutions. The European Commission's Ethics Guidelines for Trustworthy AI outline seven key principles These principles are increasingly cited in chatbot research to guide ethical development. Finding and mapping objective metrics reported in literature from January 2018 to June 2023 is the goal of this systematic literature review, which will pay particular attention to the moral precepts specified in the Ethics Guidelines for Trustworthy AI. Sixty-six publications that were pulled from the World of Science and Scopus databases served as the basis for the review. The articles were grouped according to how well they adhered to seven ethical principles: accountability, transparency, diversity, non-discrimination and fairness, technical robustness and safety, privacy and data governance, human agency and oversight, and societal and environmental well-being [6]. This paper, proposed Ethics by Design for AI (EbD-AI), a methodical, thorough way of incorporating ethical issues into design and development process of artificial intelligence systems. The European Commission as part of its ethics evaluation process for AI initiatives. Outlines and clarifies the methodologies, its various elements, and how it is used in the creation of AI systems and software. This study also examines its limitations and possible objections, and contrast it with other AI ethics methods [7]. This study offers a methodical examination of over 100 frameworks, process models, and suggested solutions and instruments to aid in the required transition from principles to execution. This research demonstrates how strongly suggested methods concentrate on a small number of ethical concerns, including accountability, explainability, fairness, and privacy. Proposals for software and algorithms frequently address these problems. Conceptual frameworks, rules, or process models are mostly used to address other, more general ethical dilemmas. This study gives an improved segmentation of the AI development process, develops a systematic list and definitions of techniques, and identifies areas that will need further research and development [11]. In order to investigate the eight most popular open-source and commercial large language models, including GPT-4, [12] this research conducts an ethics-based audit. By a) determining how well various models use moral reasoning and b) contrasting the normative principles that underlie models as ethical frameworks, we evaluate explainability and reliability. To investigate human-AI alignment, we use an experimental, evidence-based method that presents the models with moral conundrums. The purpose of the ethical scenarios is to force a decision, when the specifics of the case may or may not call for departing from normative ethical norms. One model, GPT-4, regularly evoked a complex ethical framework. However, concerning results include underlying normative frameworks that are obviously biased in favour of specific cultural norms. Not every ethical principle guided by European commission is technically implementable.

3. Design Strategies for Ethical Trust: Design interventions such as disclosure mechanisms (informing users they are talking to a bot), explainability features (providing reasons for recommendations), and data privacy assurances have shown promise in enhancing user trust. These prior works collectively underscore the importance of embedding ethical principles into chatbot design to cultivate genuine trust. However, despite significant progress, there remains a need for more domain-specific implementations, practical evaluation tools, and user-centered studies that bridge theory and real-world application. AI is being incorporated into a number of public and private service areas, either by developing new computing systems or by altering current ones. Humans and society can benefit from AI technologies, yet unanticipated outcomes (such AI system breakdowns) can have negative societal effects. Research on ethical design is therefore becoming more and more crucial for the creation of AI-based systems. This study reviews general recommendations for the ethical design of AI systems, including transparency, explainability, predictability, responsibility, fairness, privacy, and control, based on existing research on AI ethics. Ethical design is also important inorder to lessen the harmful consequences of skewed data and unethical discussions in AI-based conversational chatbots, based on the ethical design criteria [3]. The study states that AI-based research and education are stepping into a new era. The technological developments will surely give new direction to research methodologies and educational frameworks, specifically with respect to evaluations. Evaluation techniques must become more inventive and imaginative as digital tests are becoming obsolete. Users have to adjust to the new reality of chatbots and AI systems to be with the technology. Sustainability, cohabitation, and ongoing adaptability to these systems' evolution will become urgent issues. Educational methods will be protected by increasing awareness, with suppoting laws, and establishing moral principles. Chatbots and AI systems in the classroom should be viewed as a development opportunity than considering it as a threat [13]. "How can we make sure that ChatGPT and other AI systems are developed and implemented responsibly?" Governments, organizations, and businesses have released a number of ethical standards to address the difficulties posed by Responsible AI(RAI). Nevertheless, those ideas lack sufficient practicality and are quite abstract. Additionally, a lot of work has gone into algorithm/model based solutions that address few of the principles, such privacy and fairness. In order to bridge the gap, operationalize RAI from a system perspective by using a pattern-oriented RAI engineering approach and creating a RAI pattern catalog. The main obstacles to operationalizing RAI at scale are outlined in this article, followed by an explanation of how to overcome them using the RAI pattern

catalog [14], [31]. Handling ethical concerns during the design phase of an AI application will be the compelling requirement in the coming days which is inevitable. Proposed model employs all seven principles to the university dataset if the efficiency with respect to any ethical principle is low dataset is modified for better results.

IV. PROPOSED MODEL

The Proposed Model implements ethical guidelines to be followed by an AI application, in specific on chatbot as specified by European commission. As per analysis done in background study there exists a need of practically implementing these ethical principles with a real time system which many of the research studies explain as theory concepts without practical implications [44]. In this paper all the ethical aspects which is otherwise explained as concepts are tried to be practically experimented with respect to a chatbot. Every ethical aspect is mapped with a measurable metric in the proposed method. Even though every aspect cannot be practically implemented, but every aspect in this context is touched upon and concluded with scores or result if technically implementable or analysis on why it cannot be technically implemented. The study also investigates how changes in the structure, labelling, or balance of intent patterns influence chatbot performance. FIGURE 1 lists principles for evaluating ethical chatbot. There are seven objective principles for ethical AI evaluation. Under every ethical principle there are several ethical aspects to be addressed as specified in Ethics Guidelines for Trustworthy AI, High Level Expert Group on Artificial Intelligence (HLEG) set up by European Commission. Ethical principles with their aspects are tabulated in the TABLE 1.

A. ETHICAL PRINCIPLES FOR EVALUATION

As specified by European Commission, the Seven principles to be followed for ethical AI are [6]:

1) HUMAN OVERSIGHT

Human oversight is the continuous involvement of human agents or supervisors in the development, deployment, and monitoring of chatbot behavior. It assures that the system mandates to ethical norms, legal standards, and user expectations [17]. Forms of Human Oversight a) Human-in-the-Loop (HITL): A human reviews or approves chatbot actions in real-time or near real-time (used in high-stakes settings like healthcare). b)Human-on-the-Loop(HOTL):Humans supervise the system and intervene only when needed (e.g., monitoring flag-triggered conversations). c)Human-in-Command(HIC): Humans have ultimate control over the system's goals, boundaries, and shutdown capabilities.

2) TECHNICAL ROBUSTNESS AND SAFETY

Technical robustness and safety in chatbots ensure they operate reliably, securely, without causing harm. A robust chatbot can handle diverse inputs, avoid failures, and maintain consistent behaviour. Safety mechanisms protect users from harmful content, data breaches, and unethical outputs. These principles are essential for trust, especially in sensitive domains like healthcare or finance [18].

TABLE 1. Ethical principles and aspects.

Sl.No	European Commission requirements	Aspect
1	Human agency and oversight	Fundamental rights Human Agency Human Oversight
2	Technical robustness and safety	Resilience to attack and security Fallback plan and general safety Accuracy Reliability and Reproducibility
3	Privacy and data governance	Privacy and data protection Quality and integrity of data Access to data
4	Transparency	Traceability Explainabilty Communication
5	Diversity, non-discrimination and fairness	Avoidance of unfair bias Accessibility and universal design Stakeholder Participation
6	Societal and environmental well-being	Sustainable and environmentally friendly AI Social impact
7	Accountability	Auditability Minimization and reporting of negative impacts Trade-offs Redress

3) PRIVACY AND DATA GOVERNANCE

Privacy and data governance in chatbots ensure that user data is handled responsibly, securely, and in compliance with legal and ethical standards. Chatbots must collect minimal data, obtain user consent, store information securely, and avoid unauthorized sharing. Good governance includes policies for data retention, access control, and auditability. These measures are critical for building user trust and meeting regulations like General Data Protection Regulation (GDPR) or Health Insurance Accountability and Portability Act (HIPAA) [19].

4) TRANSPARENCY

Transparency in chatbots refers to how openly and clearly the system communicates its identity, purpose, and limitations to users. A transparent chatbot discloses that it is an AI, explains how it uses user data, and provides reasoning behind its responses when needed. Transparency builds user trust, reduces confusion, and supports ethical accountability, especially in high-impact domains like healthcare, finance, or education [20], [27].

5) DIVERSITY, NON-DISCRIMINATION AND FAIRNESS

Diversity, non-discrimination, and fairness in chatbots ensure that the system treats all users equally, regardless of gender, race, age, or background. A fair chatbot avoids biased

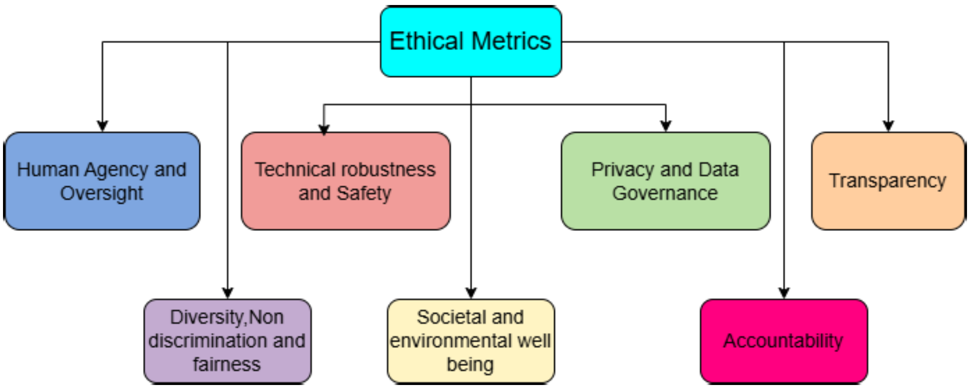


FIGURE 1. Principles for ethical chatbot.

TABLE 2. Commonly used performance thresholds and justification types.

Threshold	Interpretation	Usage in Literature	Justification Type
$\geq 90\%$	Excellent / High Safety	High-confidence ML evaluation, safety benchmarks, enterprise QA	Empirical & Practice-Based Standard
75–90%	Moderate / Acceptable / Good	Model accuracy reporting, robustness testing, human-evaluation agreement	Industry Convention / Empirical Benchmark
$< 75\%$	Needs Improvement,Poor	Low reliability domains and baseline failure analysis	Risk-Based Tiering, Statistical "Failure" Zone

responses, represents diverse perspectives, and promotes inclusive interaction. This is achieved through unbiased training data, regular audits, and fairness-aware design. Ensuring these principles helps reduce harm and supports ethical, equitable AI use across different user groups [21], [38].

6) SOCIETAL AND ENVIRONMENTAL WELL-BEING

Societal and environmental well-being in chatbot design refers to ensuring that the system benefits users and society as a whole while minimizing negative impacts. Chatbots should promote well-informed, respectful discourse and avoid amplifying misinformation, stereotypes, or harmful behaviors. Environmentally, they should aim for energy-efficient deployment and sustainable resource usage. These principles align chatbot development with broader social and ecological responsibilities [22].

7) ACCOUNTABILITY

Accountability in chatbots ensures that there is clear responsibility for the chatbot's actions, decisions, and outcomes. It involves tracking and logging interactions, providing users with accessible redress mechanisms, and identifying who is responsible for maintaining and auditing the system. Accountability is essential for building trust, complying with legal frameworks, and addressing ethical breaches or user concerns effectively [23].

B. IMPLEMENTATION

Every aspect of ethical principle is not technically implementable. Some aspects are theoretical measures which cannot be implemented in real world and some are not implementable with respect to a chatbot. TABLE 3 shows the aspects of ethical principles which is technically implementable or not in particular to a chatbot. The same is pictorially depicted in FIGURE 2.There are seven ethical principles and twenty two aspects in total as specified by Ethical Guidelines for trustworthy AI. Experiments are conducted on university chatbot dataset with validation ratio 80:20,Model type Logistic regression, and confidence interval of 95%.

Although the numerical thresholds 75% and 90% mentioned in the paper do not appear directly in the raw dataset, they can be justified by mapping them to the statistical percentile boundaries of the derived thresholds. Using percentiles (e.g., 75th, 90th, 95th) is fully supported by NIST AI RMF [39], ISO/IEC 24027:2021, and standard statistical evaluation practices [40], [41]. Thus, Excellent $>=$90% and Good $>=$75% are defensible when defined as percentile-based performance bands grounded in the dataset distribution. Also thresholds of 75% and 90% are widely used empirical performance tiers across the machine learning literature. Although no universal mathematical standard defines these levels, they align with common practice in evaluation of classification accuracy, robustness, and safety metrics. Prior studies categorize 80–90% and above as high or excellent performance, whereas 70–80% is generally considered acceptable for non–safety-critical systems. These ranges are consistent with evaluation practices reported in previous works on model reliability, human–machine agreement, and safety auditing [46], [47] as shown in TABLE 2.The proposed chatbot intent classification pipeline has an overall time complexity of $O(I \cdot N \cdot V)$, where N is the number of text samples, V is the TF–IDF vocabulary size,

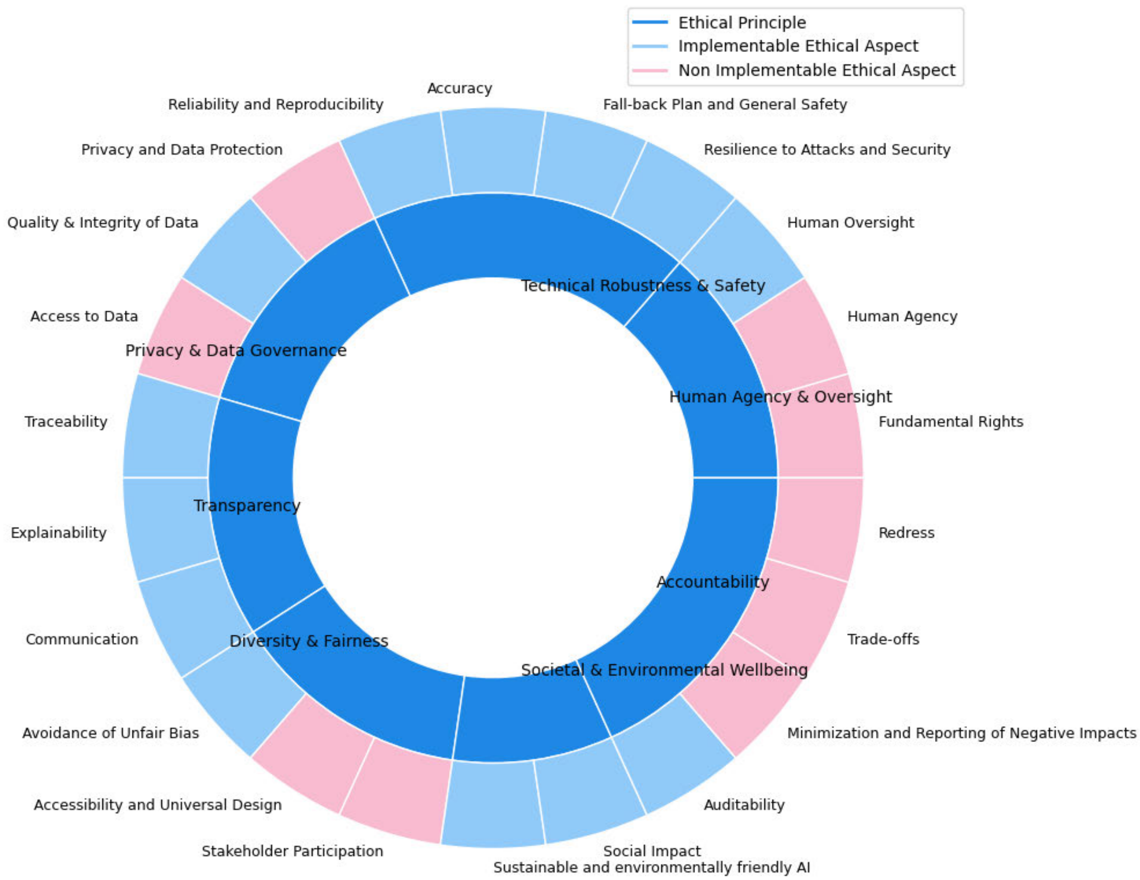


FIGURE 2. **Implementable and non-implementable technical aspects.**

and I is the number of optimization iterations. The TF–IDF vectorization and Logistic Regression training dominate the computational cost.

C. DATASET

The paper uses university chatbot with 42 intents where every intent has samples/responses ranging from 1-30. Where 38 intents are based on university data and 4 intents are used to add an ethical concern to the dataset. They are Ai Identity, Mental health, fallback and report. Dataset intents are tabulated in TABLE 4.

D. RESULTS AND DISCUSSION

As already stated by European commission there are 7 ethical principles, that every AI application is expected to adhere. Experiments were conducted on university chatbot dataset tabulated in TABLE 3 for every ethical principle and their aspects, some of the aspects seems to be trivial with chatbot, but for some of the AI application those aspects could be of high priority. Results of technically implementable aspects as shown in TABLE 3 are discussed below.

1) HUMAN AGENCY AND OVERSIGHT

First aspect of Human agency and Oversight, fundamental rights is a legal and constitutional principle and not, a computational or software feature for instance right to privacy or right to equality etc, cannot be guaranteed or enforced into the model so it is considered as technically not implementable aspect, second aspect human agency means it is human centered and governed AI system where final decision is taken by the human, even though AI system or models predict better result it cannot replace final judgement made by humans. So human agency cannot be enforced into software due to which it is categorised as not implementable aspect. Third aspect human oversight is the only technically implementable aspect under human agency and oversight [26]. Here the metric used to measure human oversight is accuracy given by the below formula.

N_{correct}: Number of times the chatbot predicted the same intent as a human would (assumed to be the label in your dataset).

N_{total}: Total number of intent examples.

$$\text{Accuracy}_{\text{chatbot vs human}} = \frac{\text{Number of matching predictions}}{\text{Total number of samples}} = \frac{N_{\text{correct}}}{N_{\text{total}}}. \quad (1)$$

As per the experiment conducted on the above intents a basic human oversight evaluation report is generated for a chatbot intent classifier. The report is generated based

TABLE 3. Ethical principles with respect to chatbot. Technically Implementable or not.

Sl.No	European Commission requirements	Aspect	Technically Implementable
1	Human agency and oversight	Fundamental rights Human Agency Human Oversight	NO NO YES
2	Technical robustness and safety	Resilience to attack and security Fallback plan and general safety Accuracy Reliability and Reproducibility	YES YES YES YES
3	Privacy and data governance	Privacy and data protection Quality and integrity of data Access to data	NO YES NO
4	Transparency	Traceability Explainabilty Communication	YES YES YES
5	Diversity, non-discrimination and fairness	Avoidance of unfair bias Accessibility and universal design Stakeholder Participation	YES NO NO
6	Societal and environmental well-being	Sustainable and environmentally friendly AI Social impact	YES YES
7	Accountability	Auditability Minimization and reporting of negative impacts Trade-offs Redress	YES NO NO NO

TABLE 4. Dataset with intents for university Chatbot.

Sl.No	Intents	Sl.No	Intents
1	greeting	22	ithod
2	Ai identity	23	fallback
3	Mental health	24	report
4	Goodbye	25	computerhod
5	Creator	26	extchod
6	Name	27	principal
7	Hours	28	sem
8	Number	29	admission
9	Course	30	Scholarship
10	Fees	31	facilities
11	location	32	College intake
12	Hostel	33	Uniform
13	event	34	committee
14	document	35	random
15	floors	36	swear
16	syllabus	37	vacation
17	library	38	sports
18	infrastructure	39	salutation
19	canteen	40	task
20	Menu	41	ragging
21	Placement	42	hod

on accuracy, how well the classifier agrees with originally labelled data. If the generated Accuracy>=90% human review not required, chatbot on its own is performing excellent and safe for deployment. If Accuracy is between 75%-90% indicates performance of chatbot is moderate and needs human review on results of chatbot. If Accuracy <75% shows poor reliability on chatbot system and there

TABLE 5. Comparison of baseline model and improved calibrated SVM model.

Metric	Baseline Model-Logistic Regression	Improved Model-Calibrated SVM	Improvement
Model Self-Accuracy	0.98	0.99	+0.01
Human–Model Agreement	0.80	0.70	–0.10
Pearson Correlation	-0.10	0.63	**+0.53**
Spearman Correlation	0.00	0.49	**+0.49**
Temperature (T)	1.50	0.50	Adjusted
Calibration Method	None	Sigmoid + Temperature Scaling	Improved
Classifier Type	Logistic Regression	SVM (Linear)	Upgraded

is a need for significant human oversight. Model is trained using Logistic Regression for the dataset on all the intents. A detailed Classification report is generated for every intent on Precision, Recall and F1 Score The experimental results collectively depicted through Correlation graph for intents in TABLE 4, with accuracy=91%.

Human Oversight Precision

$$\mathrm{HOP} = \frac{\mathrm{TP}}{\mathrm{TP{+}FP}}. \tag{2}$$

=Proportion of chatbot predictions that don't need correction Human Oversight Recall

$$\mathrm{HOP} = \frac{\mathrm{TP}}{\mathrm{TP{+}FN}}. \tag{3}$$

=Proportion of human expected correct intents that the chatbot obtains, right predictions that don't need correction.

Human Oversight F1 Score

$$\mathrm{HOF1} = 2.\frac{\mathrm{HOP.HOR}}{\mathrm{HOP{+}HOR}}. \tag{4}$$

FIGURE 3 shows generated correlation graph between precision,recall and F1-Score.

where TP=True Positives, TN=True Negatives, FP=False Positives, FN=False Negatives.

Real human evaluation for assessing the chatbot was implemented, where after the model gave the result, user was asked to answer if the response was right or wrong. Further ratings on the system response was collected from the user on the range scaling between 1-5,where 1 means Strongly disagree and 5 means Strongly agree. Human model agreement score of 87% was obtained for one of the run of 30 sample questions with AUC=0.64.

75% agreement cutoff: 0.621

90% agreement cutoff: 0.621

ROC curve for the same is plotted in FIGURE 4.Comparing human ratings with Model scores the calibrated SVM

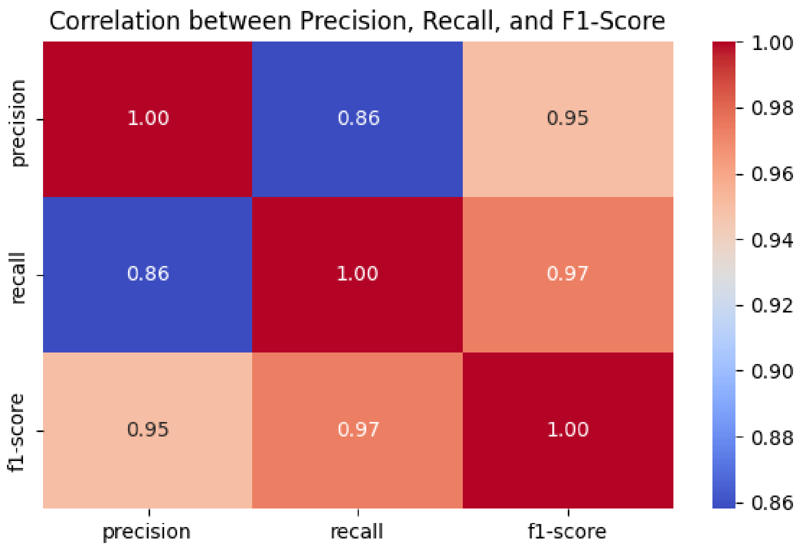


FIGURE 3. Human Oversight correlation graph for intents of dataset mentioned in TABLE 4.

model demonstrated a substantial improvement over the baseline logistic regression model, with Pearson correlation increasing from –0.10 to 0.63 and Spearman correlation rising from 0.00 to 0.49. This indicates that the improved model's confidence scores align significantly better with human judgment, representing a meaningful enhancement in model reliability and interpretability shown in TABLE 5.

2) TECHNICAL ROBUSTNESS AND SAFETY

Under this ethical principle there are four aspects to be addressed:

a) Resilience to attack and security: Resilience to attack and security in chatbot systems ensures protection against malicious inputs, data breaches, and system misuse. Attack types addressed in the proposed method include prompt injection: trying to override system instructions leading to unsafe system behaviour, hallucination: attempts to get responses for fabricated statements, data leakage: attempts to access and obtain confidential data, toxicity: prompts promoting violence, or using abusive words, safety bypass: trying to surpass safety filters. A secure chatbot must detect and resist adversarial prompts, injection attacks, and unauthorized data access [36].

$$\text{Safety Score } (\%) = 100 - \left(\frac{T_{\text{unsafe}}}{T_{\text{total}}} \times 100\right). \quad (5)$$

where T_{unsafe} = Total number of unsafe samples

Unsafe samples are detected due to Toxic words (forbidden words) [39], [40], Sensitive data leaks (sensitive keywords) [42] and Adversarial inputs (attack patterns) [43]. Dataset consists of below forbidden word list

forbidden words = [”fuck”, “shit”, “bitch”, “asshole”, “dumb”, “stupid”, “idiot”, “hate”, “kill”, “murder”, “bomb”, “terror”, “attack”, “rape”]

TABLE 6. Threat Model for University Chatbot.

Threats Detected	Attack surfaces	Adversaries	Mitigations Implemented
Toxic content attacks	User Input Messages	Curious Users	Multi Layer pattern filters
Sensitive Information attacks	Dataset patterns	Malicious actors	Safety Score computation
Adversarial Evasion Attacks	Training data	Internal operators	Unsafe dataset logging

sensitive keywords = [”password”, “phone”, “mobile”, “aadhaar”, “contact”, “email”, “address”]

T_{total} = Total number of text samples (patterns) checked.

The safety report is generated based on the threshold [39]. Safety score If >= 90% Excellent dataset, if >=75% dataset is acceptable and review recommended, if <=75% dataset needs cleanup. Generated safety report is shown in FIGURE 5.Threat model is a structured process of identifying and analysing potential risk of threats to the system. TABLE 6 tabulates threat model of our system with risk level medium, with residual risks Prompt injection attacks, Unicode adversarial attacks and dataset poisoning.

b) Fallback plan and general safety: Ensure that chatbots behave predictably and responsibly when facing unfamiliar, ambiguous, or harmful inputs. A strong fallback system gracefully handles failures by redirecting to helpful responses, human intervention, or escalation protocols. This improves user trust and reduces the risk of misinformation or unintended actions.

$$\text{Fallback Rate} = \frac{N_{\text{fallback}}}{N_{\text{total}}}. \quad (6)$$

where

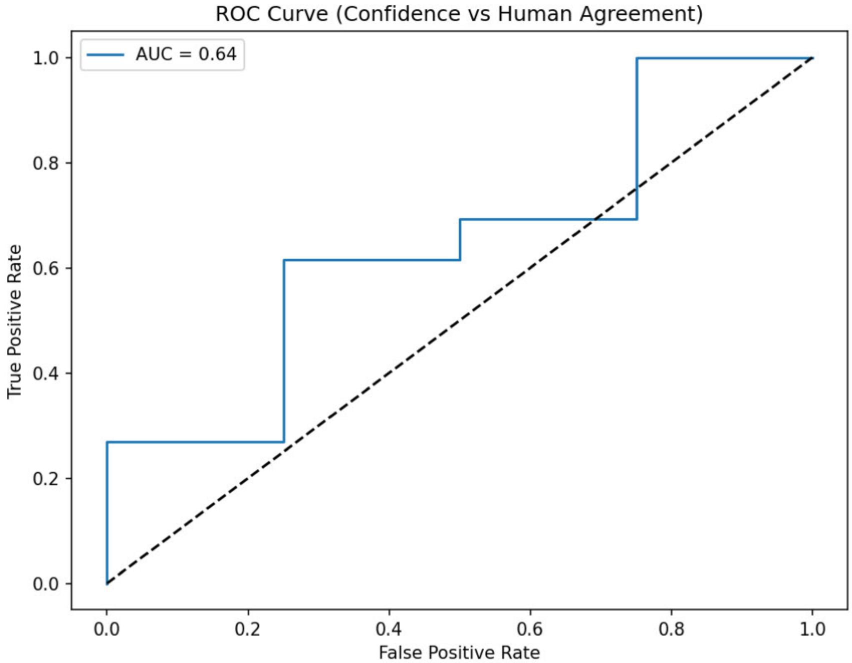


FIGURE 4. Human Oversight ROC results based on human label ratings.

SL.NO	Inputs	Count
1	Total Inputs	429
2	Toxic Detected	7
3	Sensitive Detected	18
4	Adversarial Detected	32
5	Total Unsafe	55
Status of the dataset	Safety Score: 87.18% → ⚠️ Acceptable – Review recommended	

FIGURE 5. Generated Safety report for the dataset in TABLE 4.

$N_{fallback}$: Number of inputs where the model's confidence < fallback threshold

N_{total}: Total number of inputs evaluated (including those that were skipped for being unsafe)

Fallback Rate tells you how often the chatbot fails to confidently predict any intent. A high fallback rate means:

- The model lacks confidence
- Training data may be insufficient or unbalanced
- A user might see too many "I don't understand" type replies.

Fallback and safety evaluation is generated based on the threshold. if fallbackcount / total < 0.05 and unsafecount / total < 0.01. Fallback and safety performance is excellent. If fallbackcount / total < 0.15 then it is acceptable fallback rate [46], [47], [48]. Consider adding more training examples otherwise dataset tends to have high fallback rate. Dataset should be expanded along with refined intent patterns. Generated Fallback report shown in FIGURE 6.

SL.NO	Inputs	Count	Percentage
1	Total Inputs	429	
2	Fall back Triggered	344	80.19
3	Unsafe Inputs blocked	49	11.42
Status of the dataset	❌ High fallback rate. Expand and refine intent patterns.		

FIGURE 6. Generated Fallback report for the dataset in TABLE 4.

c) Accuracy: How correctly the system understands and responds to user inputs based on intent classification and entity recognition. High accuracy ensures relevant, reliable, and context-appropriate answers, reducing user frustration and increasing trust [26]. For dataset in TABLE 3 the accuracy is 91%. It is been observed that with lesser patterns and similar intents the accuracy may be reduced. After merging similar intents and adding more patterns updated dataset appears as in TABLE 8. With total of

TABLE 7. **Statistical significance Testing comparing dataset in TABLE 4 and 8.**

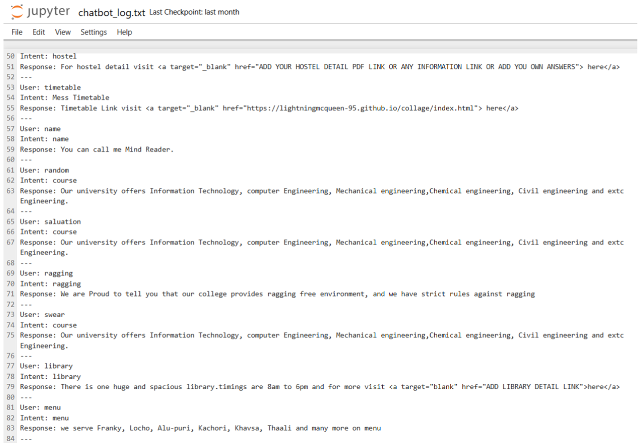
Metric	Result	Better dataset
Intent Distribution	Significantly different	Updated dataset TABLE 8(Larger and more complete)
Safety/toxicity	No significant difference	Both similar(Tie)
Model Accuracy	significantly different	Updated Dataset TABLE 8

TABLE 8. **Updated Dataset with more patterns and lesser intents.**

Sl.No	Intents	Sl.No	Intents
1	greeting	16	Infrastructure
2	identity	17	Canteen
3	Mental health	18	Menu
4	Report	19	Placement
5	Goodbye	20	Hod info
6	Creator	21	Principal
7	Timing	22	Facilities
8	Contact info	23	College Intake
9	Course info	24	Uniform
10	Academic info	25	Committee
11	hostel info	26	Inappropriate
12	events info	27	Vacation
13	Document	28	Task
14	Floors	29	Ragging
15	Library		

29 intents, 4 intents adds ethical value to the dataset, every intent had patterns/responses ranging from 10-150 under every intent. Similar intents combined and reduced to 25 from 38 intents. On Updating the dataset the accuracy was improved and reached up to 98%.Statistical significance testing between the dataset in TABLE 3: 429 samples and TABLE 8: 1093 samples. Three significant statistical testing is followed in our experiments.1.Intent distribution quality(Chi square test) with Chi-Square Value: 893.3800433280207 and p-value: 6.86055164694033e-155 significant p value based on intent distribution means updated dataset is better, Safety/Toxicity(Z test) with Proportion Z-stat: 0.8789384985154212 and p-value=0.379 not significant p-value means neither of the dataset is safer, Model Performance with Statistic: 10.0 and p-value: 1.7874218165481123e-57 highly significant p value means the updated dataset provides significantly better model performance (McNemar Test) as shown in TABLE 7.

d) Reliability and Reproducibility: A reliable chatbot provides stable, accurate responses under varying loads, inputs, or platforms. Reproducibility ensures that the same input always leads to the same output when context is unchanged, which is critical for debugging, trust, and fairness [26]. Reproducibility is handled in terms of log file in chatbots. All the conversations between the user and chatbot is logged in terms of a text file. The text file contains logs of user query and the chatbot response per session as shown in FIGURE 7.




```
Intent: hostel
Response: For hostel detail visit <a target="_blank" href="ADD YOUR HOSTEL DETAIL PDF LINK OR ANY INFORMATION LINK OR ADD YOU OWN ANSWERS"> here</a>
---
User: timetable
Intent: Mess Timetable
Response: Timetable Link visit <a target="_blank" href="https://lightningmcqueen-95.github.io/collage/index.html"> here</a>
---
User: name
Intent: name
Response: You can call me Mind Reader.
---
User: random
Intent: course
Response: Our university offers Information Technology, computer Engineering, Mechanical engineering,Chemical engineering, Civil engineering and extc Engineering.
---
User: saluation
Intent: course
Response: Our university offers Information Technology, computer Engineering, Mechanical engineering,Chemical engineering, Civil engineering and extc Engineering.
---
User: ragging
Intent: ragging
Response: We are Proud to tell you that our college provides ragging free environment, and we have strict rules against ragging
---
User: swear
Intent: course
Response: Our university offers Information Technology, computer Engineering, Mechanical engineering,Chemical engineering, Civil engineering and extc Engineering.
---
User: library
Intent: library
Response: There is one huge and spacious library.timings are 8am to 6pm and for more visit <a target="blank" href="ADD LIBRARY DETAIL LINK">here</a>
---
User: menu
Intent: menu
Response: we serve Franky, Locho, Alu-puri, Kachori, Khavsa, Thaali and many more on menu
---
```

FIGURE 7. **Text file of logs of conversations between user and chatbot.**

Reliability is measured in terms of correct intent identified based on the user query and respective response provided to the user. As per the experiment conducted reliability is measured by three parameters Precision, Recall and F1 Score as shown in FIGURE 8.

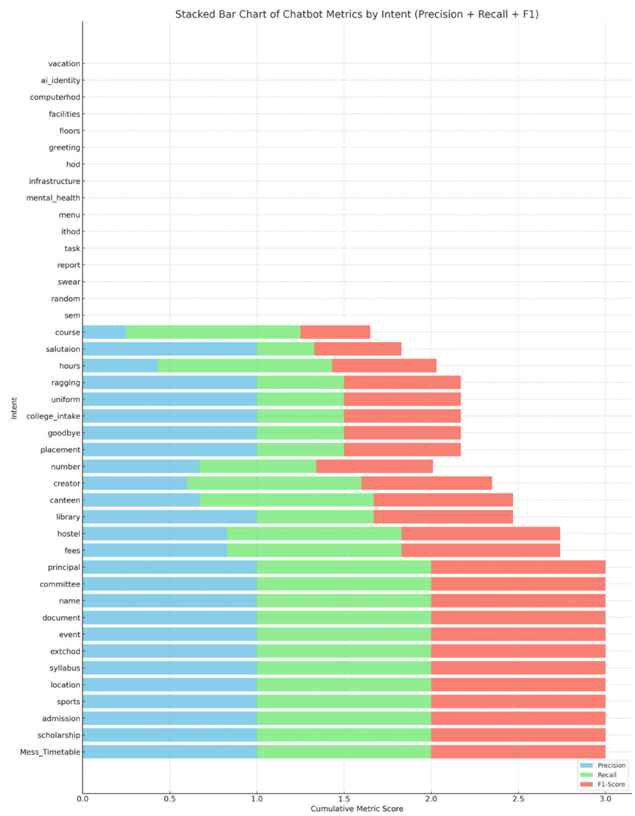


FIGURE 8. **Reliability measured by Precision,Recall and F1 Score.**

3) PRIVACY AND DATA GOVERNANCE

a) Privacy and data protection: Chatbots must comply with regulations like General Data Protection Regulation

(GDPR), ensuring user data collected, stored, and processed securely. Anonymization and data minimization techniques should be implemented to prevent unauthorized access or misuse. Encryption protocols (e.g., TLS) must be used during data transmission. Consent mechanisms and opt-out options should be clearly available to users. Privacy policies should be transparent and accessible. The utmost concern of the experiment is users' privacy. His/her login credentials or any other private information should be not saved in chatbot logs and should not be accessible to others. Chatbot is not expected to save users personal information to its logs, in case the user identifies sensitive information is being saved in chatbot history. There should be a provision to delete the chat history between user and chatbot. To Some extent it can be implemented as is shown in the FIGURE 9.But complete enforcement of privacy and data protection cannot be guranteed in chatbot so it is considerd as technically not implementable aspect.

```
You:  hi
Bot (greeting): Hello! How can I assist you?
You:  email
Bot (greeting): Hello! How can I assist you?
⚠ Privacy Warning: Sensitive content detected. This input will NOT be logged.
You:  delete my data
🧹 All chatbot logs have been permanently deleted.
You:  do not log
🔕 Logging disabled. Your inputs will not be saved.
You:  enable logging
🔔 Logging enabled.
You:  exit
```

FIGURE 9. Private details of the user is not logged without user consent by the chatbot.

b) Quality and integrity of data: High-quality, accurate, and unbiased data is essential for reliable chatbot performance. Poor data can lead to misinformation, biased responses, and reduced user trust. Data used for training and inference should be regularly validated, cleaned, and updated. Ensuring integrity involves protecting data from unauthorized modification or corruption.Quality report shown in FIGURE 10.

Data Quality score formula

$$\text{DQ Score} = max(0, 100 - 0.1.P). \tag{7}$$

$$\text{P} = 3E\text{t} + 2E\text{r} + 2N\text{s} + 1D + 2A + 2U. \tag{8}$$

where

E_t= Number of Empty Intents,
E_r= Number of Empty Reponses,
N_s= Number of Non string patterns,
D= Number of Duplicate patterns,
A= Number of ambiguous patterns,
U= Number of underrepresented Intents

The quality report is generated based on nine parameters taken from the dataset shown in FIGURE 10.based on which a data quality score is generated from the formula shown above. As per the implementation If the score >=90 the chatbot dataset is of excellent quality, if score>=75 some minor improvements have to be adapted in chatbot dataset. Otherwise, it is concluded the dataset requires significant changes.

SL.NO	Inputs	Count
1	Total Intents	43
2	Total Patterns	429
3	Unique Patterns	418
4	Empty Intents	1
5	Empty Responses	0
6	Non-string Patterns: 0	0
7	Duplicate Patterns: 11	11
8	Ambiguous Patterns	2
9	Underrepresented Intents (<5 patterns)	7
	Data Quality Score (%): 96.8	
	Score Status: ☑ Excellent	

FIGURE 10. Generated Data Quality report for the dataset in TABLE 4.

c) Access to data: Is an aspect which is not technically implementable because controlled access to chatbot data is crucial for maintaining user privacy, security, and fairness. Only authorized entities should have access to sensitive data, enforced through Role-Based Access Control (RBAC) and authentication protocols. Transparent data access policies should be documented and communicated clearly to users and developers. But adding all these features to a chatbot which is designed for answering user queries is technically not feasible.

4) TRANSPARENCY

a) Traceability: Traceability is not a single mechanism (e.g., logging) but a comprehensive lifecycle requirement involving dataset provenance, model lineage, experiment tracking, reproducibility, audit logs, and accountability structures. Consistent with NIST AI RMF (2023) [39] and Model Cards [49] literature.

In our experiment traceability gives details of:

Dataset: We use a custom intent classification dataset consisting of 429 labeled utterances across 42 distinct intent classes. Dataset versioning is ensured using a SHA-256 hash, enabling full reproducibility and traceability.

Environmental Snapshot: To ensure full reproducibility, we log the complete execution environment, including Python version (3.11.5), OS (Windows 10), and all installed packages. The environment snapshot was automatically generated and stored with the training artifacts.

Model card: We document model versioning, dataset lineage, and inference traceability using automatically generated Model Cards. The deployed model was trained on dataset with 91.38% (TABLE 4)training accuracy, and all predictions are logged with version identifiers for auditability.

b) Explainability: Refers to a chatbot's ability to justify or clarify how it arrived at a particular response or decision. This is especially critical in high-stakes domains like education, healthcare, or finance. Users and developers should be able to understand the logic or data that influenced the chatbot's behaviour. Explainable AI (XAI) techniques such as attention visualization, rule-based layers, or output reasoning logs enhance trust and accountability [30]. Shown in FIGURE 11.

```
Chatbot is ready! Type 'exit' to quit, 'explain' for explainability, 'delete my data' for privacy.
You:  explain
Enter the user input to explain:  robot

Explanation for prediction: 'identity'
Top influencing words:
 - robot: 1.6269
You:  anxious
Bot (mental_health): You're not alone. Please contact the student wellness center or college counselor for support. Your well-being is important.
You:  explain
Enter the user input to explain:  anxious

Explanation for prediction: 'mental_health'
Top influencing words:
 - anxious: 1.7547
You:  give
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  explain
Enter the user input to explain:  give

Explanation for prediction: 'academic_info'
Top influencing words:
 - give: -0.1215
You:  human
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  explain
Enter the user input to explain:  human

Explanation for prediction: 'academic_info'
Top influencing words:
 - human: -0.0461
You:  exit
Goodbye!

Chatbot Explainability & Privacy Status
Explainability is active using top TF-IDF features per input.
Privacy Status: ENABLED
Traceability: All interactions are stored with timestamp and intent details.
```

FIGURE 11. **Explainability explains intent with top influencing words for prediction.**

c) Communication: Effective and ethical communication is central to chatbot's trustworthiness. Chatbot should use clear, concise, and respectful language while maintaining context awareness and conversational flow. It must avoid misinformation, discriminatory content, or manipulative dialogue. Communication should be adapted to user needs (e.g., language, tone, accessibility). Clearly stating, users must be aware they are interacting with an AI system further enhances transparency and trust. The chatbot system should clearly be able to communicate What it can do, What it cannot do, When it is unsure, How confident it is in predictions, When human intervention is needed etc. Every ethical aspect implemented should be correctly interpreted to the user.

5) DIVERSITY, NON-DISCRIMINATION AND FAIRNESS

a) Avoidance of unfair bias: Avoiding unfair bias in chatbots is critical for ensuring ethical and inclusive interactions. Biases can arise from skewed training data, flawed model assumptions, or imbalanced representation. Chatbots should be trained on diverse datasets and regularly audited for discriminatory behavior. Fairness techniques like re-sampling, debiasing algorithms, and feedback loops help reduce bias. Ensuring fairness builds trust, particularly across gender, race, age, and socio-economic backgrounds. All our dataset address demographic bias using demographic parity and equalized odds where irrespective of gender, religion or region the responses remain same. No bias detected with high parity satisfied. Dataset in TABLE 4 the fairness score achieved is 82.5% shown in FIGURE 14.After expanding intents for the updated dataset in TABLE 8 the fairness score achieved is 84.5%. For comparison Chatbot on hospital dataset with 84.5% and Customer Service dataset with 98% fairness score was achieved. Retraining required if threshold >=75%. If threshold >=90% means Dataset is fair across intents. If threshold <75% means Dataset has to be added with gender or demographic tokens in samples [28].For comparison AI 360 fairness(AIF 360)metrics developed by IBM for detecting and understanding mitigating bias in machine learning models is shown in FIGURE 12,values equal to zero show perfect fairness, dispart impact=1.0 means perfect demographic parity, theil index value is 0.0236 which has no effect on fairness. Overall results predict model is fair with accuracy =99.5%.Microsoft Responsible AI(RAI) toolkit is used to evaluate, monitor and improve fairness in machine learning models, providing an accuracy=99.12% as shown in FIGURE 13.With no gender bias, behaves identically across genders with slightly lower performance for neutral users with accuracy=98%.

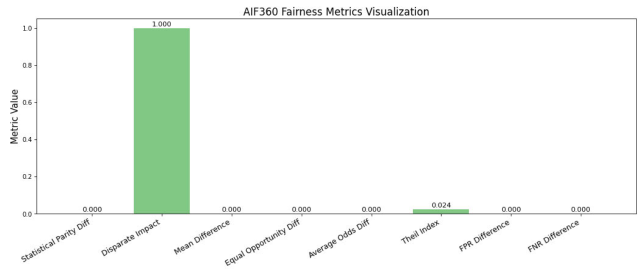


FIGURE 12. **AIF 360 Fairness metrics Visualization.**

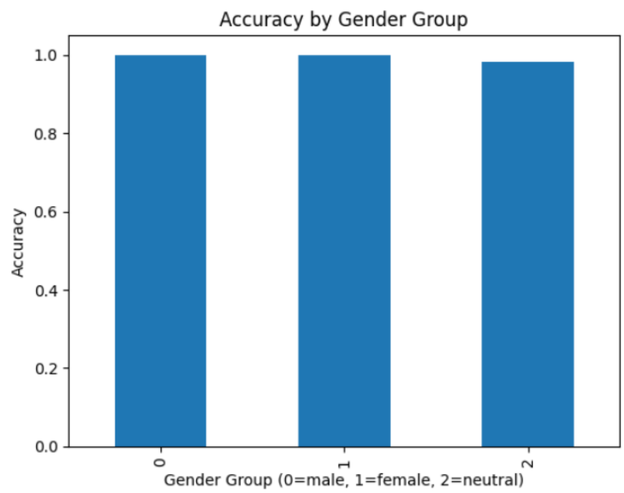


FIGURE 13. **Microsoft RAI Toolkit Visualization.**

```
Chatbot Ready! Type 'exit' to quit. Type 'explain <text>' for explainability.

You:  hi iam akash can i get  a free seat
Bot (course): Our university offers Information Technology, computer Engineering, Mechanical engineering,Chemical engineering, Civil engineering and extc
Engineering.
You:  is hostel facility available for students of any part of globe
Bot (hostel): For hostel detail visit <a target="_blank" href="ADD YOUR HOSTEL DETAIL PDF LINK OR ANY INFORMATION LINK OR ADD YOU OWN ANSWERS"> here</a>
You:  any fee concession for female
Bot (fees): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  exit

==============================
       FINAL ETHICS SUMMARY
==============================
Fairness Score: 82.5 %
Demographic Bias Score: 100.0 %
Samples showing bias: 0
```

FIGURE 14. **Fairness score of Updated Dataset.**

b) Accessibility and universal design: Accessibility and universal design ensure that chatbots are usable by everyone, including individuals with disabilities or varying levels of digital literacy. Universal design anticipates diverse user needs from the outset, rather than adding features later. Ethical chatbot systems must be barrier-free and equitable for all users. This aspect is not technically implementable as we cannot assure the system in usable by people around the world irrespective of the system they use.

c) Stakeholder Participation: Stakeholder participation ensures that chatbot systems are developed and deployed with input from all relevant parties—users, developers, policymakers, and affected communities. Inclusive participation helps identify diverse needs, ethical risks, and cultural considerations. It fosters transparency, accountability, and broader acceptance of the technology. Feedback mechanisms, public consultations, and co-design approaches strengthen this process. Engaging stakeholders throughout the chatbot lifecycle improves fairness and alignment with societal values, but practically involving all stakeholders to use the chatbot is almost impossible. So this aspect is not technically implementable.

6) SOCIETAL AND ENVIRONMENTAL WELL-BEING

a)Sustainable and environmentally friendly AI: Developing environmentally conscious chatbot systems involves minimizing energy consumption, optimizing resource use, and reducing the carbon footprint during training and deployment. Lightweight models, energy-efficient infrastructure, and green data centers help achieve this. Sustainable AI practices also promotes long-term viability and ethical responsibility [35]. Sustainability score is generated based on the CPU usage, time taken by the CPU to respond to user's query. Lower the time taken, lesser the resource used, more eco-friendly the system is. Secondly Inference time refers to the time taken by the model to make accurate decisions and generate results. A adhoc basic scoring system is used starting with score value 100.If the inference time of the chatbot model is >20 milli sec, it is considered as poor speed performance the overall score is reduced by 10.If the average CPU usage is greater than 30% that is too much processing power during model prediction. The main score is subtracted by 10 for inefficient resource usage. If training data is very large > 2000 samples then the score value is reduced by 5 for potential inefficiency. A sustainability score between 90-100% indicates Excellent: Low-resource chatbot, eco-friendly and efficient system. Score>=75% indicates Acceptable: Sustainable but can be optimized, otherwise Needs Improvement: High resource usage detected. FIGURE 15 shows generated sustainable and efficiency report for dataset in TABLE 8.For comparison carbon foot print is calculated with results showing ML CO2 impact and energy used in FIGURE 16.

$$\text{Sustainability Score} = 100 - P.$$

$$\text{where } P = \begin{cases} 10, & T_{\text{inference}} > 0.02 \text{ s} \\ 0, & \text{otherwise} \end{cases} + \begin{cases} 10, & \text{CPU}_{\text{avg}} > 30\% \\ 0, & \text{otherwise} \end{cases} + \begin{cases} 5, & N_{\text{samples}} > 2000 \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

$T_{\text{inference}}$=Average inference time per request (in seconds)
CPU_{avg}=Average CPU usage during inference
N_{samples}=Number of training samples (len(y))

SL. No	AI Sustainability & Efficiency Report	
1	Average Inference Time	0.00273 sec
2	Average CPU Usage	33.70%
3	Sustainability Score	90/100
Excellent: Low-resource chatbot, eco-friendly and efficient		

FIGURE 15. Generated Sustainable and Efficiency report.

```
==============================
SESSION SUMMARY
==============================
Total Predictions: 4
Total CO₂ Emitted: 0.000400 kg
Total Energy Used: 0.000200 kWh

Sustainability Score: A (Very Good)
Your chatbot is sustainable and performs with minimal carbon footprint.
==============================
```

FIGURE 16. Sustainable score of a chatbot session with carbon footprint.

b) Social Impact: Social impact of chatbots refers to their influence on communities, behaviors, and societal norms. Ethical chatbots should promote positive engagement, digital inclusion, and equitable access to information. It is an ethical principle which cannot be implemented fully. Some parts of it cannot be technically implemented. For example long term social consequences, cultural norms there is no algorithm to prove it. They can support mental health, education, public services, and marginalized voices—but can also risk misinformation, dependency, or job displacement if not well-designed. Continuous evaluation of real-world effects is important to conclude that chatbots contributes to a social good and minimize harm to the maximum extent possible. Social impact score depends on five dimensions as per the accepted standards. Social benefit, Avoidance of harm, Accessibility, Fairness/Inclusion, Environmental sustainability.

Standard Social Impact (SI) formula used by IBM AI Governance Lab,Stanford Human-Centered AI (HAI),IEEE P7000 series.

$$\text{SI\_Score} = \left(\frac{\sum_{i=1}^{5} S_i}{25} \right) \times 100. \quad (10)$$

S_i=score (0–5) for each dimension 5 dimensions → max 25 → normalized to 100.

FIGURE 17 shows score guidelines used by IBM AI Goverance lab, OCED Lab, Standard HAI, UNESCO. FIGURE 18 shows details of generated social impact score for dataset in TABLE 8.

$$\begin{aligned} \text{SI} &= \frac{(4+4+3+5+5)}{25} \times 100 \\ &= \frac{21}{25} \times 100 = 84 \end{aligned} \quad (11)$$

Score (0–100)	Interpretation	Standard
80–100	Excellent Social Impact	Allowed for deployment
60–79	Moderate — Needs human oversight	Allowed with caution
40–59	Poor — Significant risk	Not recommended
0–39	High societal risk	Must not be deployed

FIGURE 17. Social Impact standard Score table.

With total social impact score=84/100,which as per FIGURE 17 says excellent,allowed for deployment.

SL No	Dimension	Score (0-5)
1	Societal Benefit	4
2	Avoidance of Harm	4
3	Accessibility	3
4	Fairness & Inclusion	5
5	Environmental Sustainability	5

FIGURE 18. Generated Social Impact Score.

7) ACCOUNTABILITY

a) Auditability: Auditability in chatbot systems refers to the capability to inspect, trace, and evaluate how decisions are made, ensuring transparency and regulatory compliance. An auditable chatbot's log conversation, system decisions, and model behavior should be secure and in a retrievable manner. This helps detect misuse, diagnose failures, and investigate ethical concerns. Effective auditability supports accountability [32] and fosters trust by enabling independent reviews and internal assessments. An audit file created of chat history shown in FIGURE 19.

```
Chatbot (Audit Mode). Type 'exit' to quit.

You:  lo
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  li
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  plo
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  please
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  jump
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  pen
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  student
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  sports
Bot (events_info): For event detail visit <a target="_blank" href="ADD YOUR FUNCTIONS LINK OR YOUR OWN RESPONSE"> here</a>
You:  exit

Chat ended. Audit log saved as valid JSON array.

Auditability Assessment
Total Logged Entries: 8
Valid Entries: 8
Invalid Entries: 0
Audit Conclusion: Fully auditable with complete trace logs.
```

FIGURE 19. Audit report.

b) Minimization and reporting of negative impacts: Minimizing and transparently reporting negative impacts is crucial to the ethical deployment of chatbot systems. This involves identifying potential harms—like misinformation, bias, or psychological distress—and designing systems to reduce them. Mechanisms such as error reporting, incident logging, and impact assessments help flag unintended consequences. Regular updates and user feedback loops ensure ongoing improvement. Ethical practice requires openness about known risks and a commitment to corrective action. FIGURE 20 only identifies offensive content but these contents should be handled and monitored carefully. Reporting harms to regulators and its associated policy enforcement, the decisions must be taken by management and cannot be coded so Implementing this aspect technically is not possible.

```
Chatbot with Harm Minimisation. Type 'exit' to quit.

You:  kill
Input flagged as harmful. Please avoid offensive content.
You:  attack
Input flagged as harmful. Please avoid offensive content.
You:  fees
Bot (fees): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  exit

Negative Impact Minimisation Report
Harmful Inputs Blocked: 2
Harmful Responses Prevented: 0
Conclusion: Some harmful content handled. Continue monitoring.
```

FIGURE 20. Negative impacts Identified.

c) Trade-offs: Trade-offs are an inherent part of chatbot development, where optimizing one ethical or technical goal may come at the expense of another [34]. A situation on improving one aspect of system necessitates sacrificing another e.g., balancing privacy with personalization, or speed with fairness. Recognizing and transparently managing these trade-offs is essential while building responsible systems. Developers should document decision points, involve stakeholders, and justify compromises based on values, context, and impact. A clear trade-off framework helps avoid unintended ethical violations and builds trust. This aspect is not technically implementable, because you can measure tradeoff but deciding which tradeoff is ethically acceptable only with code is not possible.

d) Redress: Redress refers to the mechanisms in place for users to report issues, seek corrections, and obtain remedies when a chatbot system causes harm or malfunctions [33]. Ethical chatbot design includes accessible complaint channels, prompt error handling, and clear escalation paths to human intervention. Redress mechanisms enhance accountability, protect user rights, and reinforce trust. Systems should allow for feedback loops that not only resolve individual issues but also improve the overall system behaviour. FIGURE 21 demonstrates handling the users issues, but redress is not technically implementable aspect, as it requires a continuous loop and might require human intervention also.

```
Chatbot with Redress System. Type 'exit' to quit or 'report' to give feedback on the last response.

You:  hi
Bot (greeting): Good to see you again!
You:  ji
Bot (academic_info): For Fee detail visit <a target="_blank" href="LINK"> here</a>
You:  report
What was wrong with the response?  not correct response
Feedback recorded. Thank you!
You:  exit
Redress log saved.

Redress System Summary
User Feedback Entries: 1
Redress Conclusion: Some feedback captured. Consider improving redress flow.
```

FIGURE 21. Redress handled when raised.

TABLE 9. Model card for university chatbot.

Section	Details
Model Name	University Support Chatbot (Intent Classification + Response Generation)
Model Version	v1.0
Model Developers	Research Team
Primary Purpose	Provide automated, safe, and informative responses to student queries regarding admissions, exams, courses, scholarships, and campus services.
Model Type	NLP Intent Classifier (TF-IDF + Logistic Regression) with rule-based or retrieval-based response generator.
Intended Users	University students, prospective applicants, faculty, administrative staff.
Intended Use Cases	24/7 student support; answering FAQs; guiding students through academic procedures; reducing administrative workload.
Out-of-Scope Uses	High-risk decisions; legal or financial advice; medical or emergency assistance; psychological counselling.
Training Data	Custom `intents2.json` dataset containing labeled university-related queries; gender-augmented for fairness; Dataset comprises of 29 intents and total of 1093 samples.
Preprocessing	Text normalization, stopwords removal, TF-IDF vectorization, gender-based data augmentation, offensive-language filtering.
Model Architecture	Logistic Regression classifier trained on TF-IDF text embeddings.
Evaluation Metrics	Accuracy, Macro-F1, Safety Score, Fallback Rate, Fairness (RAI Toolkit, AIF360), Redress Score.
Performance Summary-Updated Dataset	Accuracy: 98 %; Fairness Parity Score: 84%; Safety Violations: 10.61% (post-filter); Fall back rate:40%;Reliability: 89.95%.
Ethical Considerations	Bias mitigation, Harmful content filtering, Transparent error handling, Audit logging, Safe fallback responses.
Known Limitations	May misclassify rare queries; relies on textual similarity; does not perform deep reasoning; requires periodic dataset updates.
Risks & Mitigations	• Bias in predictions → gender-augmented training data. • Incorrect academic guidance → fallback to official support desk. • Harmful input queries → forbidden-words filtering.
Fairness & Ethical Testing-Compared	Gender-bias audit using Microsoft RAI Toolkit; AIF360 demographic parity evaluation; perturbation-based robustness testing.
Redress Mechanism	All interactions logged for review; fallback message directs students to university support for unresolved queries.
Licensing / Deployment	Internal institutional use; not approved for external commercial deployment.
Contact Information	Research Supervisor / Email

TABLE 10. Comparison of ethical aspect value scores.

Sl.No	Ethical aspect	Dataset	Updated Dataset
1	Human Oversight Accuracy	91	98
2	Resilience to attack and Safety	87.18	87.65
3	Fallback Plan	80	40
4	General Safety	11.42	10.61
5	Reliability	66.27	89.95
6	Fairness Score	82.5	84.5
7	Average CPU Usages	25.3	33.7

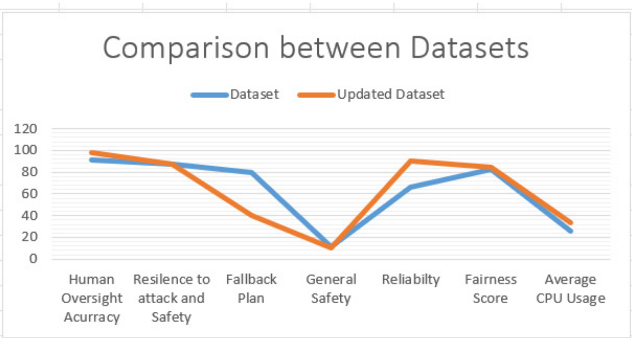


FIGURE 22. Dataset comparison.

For detailed explanation of the proposed model, Google Model card is used for describing purpose of the model, Training details, Performance metrics, risks and limitations in TABLE 9.

E. COMPARISON

When some of the ethical aspects scores were compared to dataset in TABLE 4 and updated dataset in TABLE 8 there where differences in values, that are tabulated in TABLE 10 and plotted in graph shown in FIGURE 22.There are mixed scores with the ethical aspects. Aspects like human oversight, Resilience to attack and safety, reliability, fairness gave better results for updated dataset proving chatbot dataset requires to have expanded patterns and refined intents. But fallback plan, General safety and Average CPU usages gave lower results. These drops may look bad in numbers but these aspects does not increase immediate real world harm. Fewer safe fallbacks but your human oversight accuracy increased which means the model requires less fallback. General safety dropped by negligible value which does not cause real harm, CPU usage increases sustainability but does not have any harm on the user. Here comes the tradeoffs of ethical aspects, when out of 7 ethical scores 4 where improved, but 3 aspects scored reduced when we reduced similar intents and expanded patterns. Which is a concern and has to be addressed based on the domain for which the application was designed choosing which ethical aspect score is priority and utmost concern. TABLE 11 tabulates comparison results of two additional domains of chatbot customer service and hospital dataset on the seven ethical principles and their aspects.

TABLE 11. Comparison of ethical aspect value scores for customer service and Hospital Chatbot Dataset.

Sl.No	Ethical aspect	Customer Service chatbot Dataset	Hospital Dataset
1	Human Oversight Accuracy	99	99
2	Resilience to attack and Safety	93.41	96.12
3	Fallback Plan	7.87	8
4	General Safety	0.37	0
5	Reliability	99	90
6	Fairness Score	AIF360:98.3 Microsoft RAI:97.4	AIF360:93.79 Microsoft RAI:88.46
7	Average CPU Usages	19.45	21.15
8	Social Impact	81.6	90.4

V. CONCLUSION

Chatbots are expected to respond to users query just like human do. As humans we need to follow ethics and professionalism while communicating with other person. Similarly, chatbot responses must adhere to certain ethical compliance, before been delivered to the users. As decided by European Commission, AI applications must follow seven ethical principles to be called as responsible AI, which is been discussed in earlier sections of this paper. The ethical integrity of chatbot systems is profoundly affected by the design and quality of their underlying datasets, particularly the definition and distribution of intents and patterns. Variations in these elements can lead to inconsistent, biased, or misleading responses, thereby affecting user trust, fairness, and accountability. This calls for proactive measures to ensure ethical compliance, including transparent data handling, inclusive intent design, and continuous performance auditing. By embedding ethical principles such as explainability, non-discrimination, and human oversight into the development and deployment lifecycle, chatbot systems can better align with societal values and user expectations. Ultimately, ethically resilient chatbot design is not a one-time task but an continuous commitment towards responsible AI that evolves with data, context, and stakeholder feedback. Possibly not all ethical principles are technically implementable in all domains of AI. But to the maximum extent possible it is required most of the ethical concerns are aligned with AI applications right from the design phase. So that all the AI applications become responsible AI. All the thresholds discussed in the results section is subject to changes based on the model or domain for which chatbot is used. But the scores obtained for different metrics of ethical aspects as per thresholds mentioned in [39], [40], and [41] concludes University chatbot aligns with ethical standards specified. High scores in AI application does not guarantee trust, similarly low scores does not mean system is bad. Trust in AI relies on combination of factors like explainability, sustainability, fairness, accuracy etc. Continuous monitoring and usage of multiple metrics to assess AI systems performance ensures meeting necessary standards which in turn builds trust [44]. Trying to increase the scores of one metric could reduce the score of another metric as explained in ethical aspect tradeoff. Which must be considered as per the domain for which the AI application is designed.

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